# SOLUTIONS: STABILITY AND CONTROL OF NONLINEAR SYSTEMS MASTER IN CONTROL

1. Exercise

a) The vector field $f(x_1,x_2)$ is smooth (i.e. $\mathscr{C}^\infty$) with respect to both variables. Hence it is a locally Lipschitz function and by Caratheodory's theorem solutions exist and are unique (over their maximal time-interval of definition). [ 2 ]

b) The nullclines of the vector-field can be computed as shown below:

$$\mathscr{N}_1 = \{x : x_2 = x_1 - \sin(2x_1)\}.$$

This is a function which intersects the $x_1$ axis in 0. For the second nullcline we have:

$$\mathscr{N}_2 = \{x : x_1 = -x_2 + \sin(2x_2)\}.$$

[ 2 ]

c) There are 4 regions delimited by the nullclines. The direction of the vector-field in each one of them is shown in Fig. 1.1. [ 2 ]

d) Nullclines intersect in 1 point, $[0,0]'$ which is the only equilibrium of the system. To linearize the dynamics we first compute the Jacobian matrix $\frac{\partial f}{\partial x}$ at an arbitary value of $x$.

$$\frac{\partial f}{\partial x}(x) = \begin{bmatrix} -1+2\cos(2x_1) & 1 \\ -1 & -1+2\cos(2x_2) \end{bmatrix}.$$

At the equilibrium $x_e = [0,0]'$ we see that:

$$\frac{\partial f}{\partial x}(x_e) = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Its characteristic polynomial is $\chi_1(s) = s^2 - 2s + 2$, whose roots are $-1 \pm j$. Two complex conjugate eigenvalues of positive real part correspond to the phase portrait of an unstable focus (hyperbolic). [ 3 ]

e) To verify forward invariance of the discs we let $g(x) := x_1^2 + x_2^2 - M$ and compute:

$$\begin{aligned} \frac{\partial g}{\partial x}(x) f(x) &= 2x_1(-x_1 + \sin(2x_1) + x_2) + 2x_2(-x_2 + \sin(2x_2) - x_1) \\ &= -2x_1^2 - 2x_2^2 + 2x_1\sin(2x_1) + 2x_2\sin(2x_2) \\ &\leq -(x_1^2 + x_2^2) - |x_1|^2 + 2|x_1| - |x_2|^2 + 2|x_2| \\ &= -|x|^2 - (|x_1|-1)^2 - (|x_2|-1)^2 + 2 \leq -|x|^2 + 2. \end{aligned}$$

Hence for all $M \geq 2$, the set $D_M$ is forward invariant. [ 4 ]

f) Because the equilibrium in 0 is an unstable focus, there exists a sufficiently small $\varepsilon > 0$ such that the set $\{x : \varepsilon <= |x|^2 \leq M\}$ is forward invariant, for all $M \geq 2$. This set is forward invariant and does not contain equilibria. [ 3 ]

g) Every initial condition with $x_0 \neq 0$ belongs to a set of this kind, and by Poincaré-Bendixson criterion its $\omega$-limit set is a periodic solution. [ 2 ]

h) A sketch of the global phase portrait is shown in Fig. 1.2. [ 2 ]

Answers

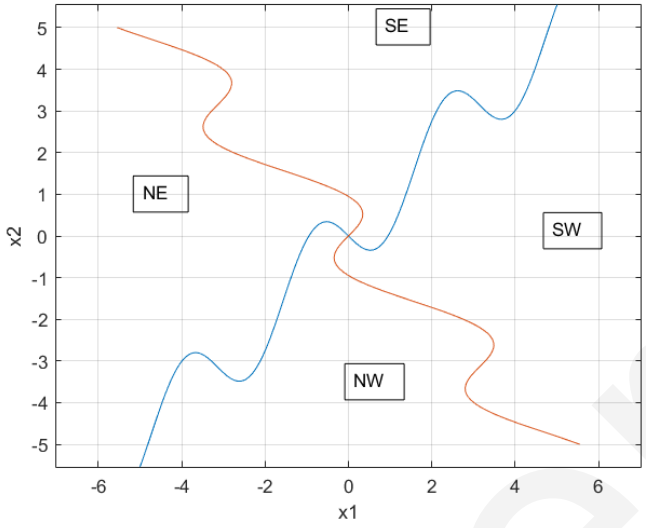



Figure 1.1 Nullclines and vector-field orientations

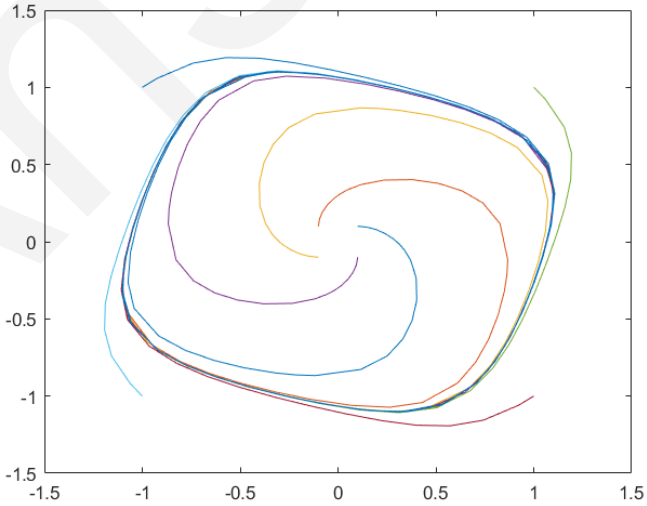



Figure 1.2 Global phase portrait

Answers

2. Exercise

a) We consider $x = [z, \dot{z}]' = [x_1, x_2]'$ as the state variable. Accordingly we see:

$$\dot{x} = \begin{bmatrix} x_2 \\ -x_2(x_1^2 + x_2^2) - x_1^5 + u \end{bmatrix}$$
$$y = x_2.$$

[ 3 ]

b) Notice that this is a control-affine system $\dot{x} = f(x) + g(x)u$:

$$f(x) = \begin{bmatrix} x_2 \\ -x_2(x_1^2 + x_2^2) - x_1^5 \end{bmatrix} \qquad g(x) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

[ 2 ]

c) To show passivity of a control-affine system we need to find a storage function $S(x)$ such that:

$$L_g S(x) = y = x_2 \qquad L_f S(x) \leq 0.$$

In particular then, we choose:

$$\frac{\partial S}{\partial x_2}(x) = x_2.$$

This implies:

$$S(x) = \frac{x_2^2}{2} + q(x_1)$$

for some suitable function $q$. Taking derivatives of $S$ yields:

$$\dot{S} = q'(x_1)x_2 - x_2^2(x_1^2 + x_2^2) - x_2 x_1^5.$$

Letting $q'(x_1) = x_1^5$ yields:

$$\dot{S} = -x_2^2(x_1^2 + x_2^2) \leq 0$$

Hence, the storage function:

$$S(x) = \frac{x_2^2}{2} + \frac{x_1^6}{6}$$

can be used to prove passivity. [ 4 ]

d) Letting $V(x) = S(x)$ we see that $V$ is positive definite (sum of two non-negative terms, and zero only when both are zero, viz. at the origin). On the other hand $V$ is also radially unbounded. In fact:

$$V(x) \leq M \Rightarrow x_2^2 \leq 2M \text{ and } x_1^6 \leq 6M \Rightarrow |x| \text{ is bounded}$$

[ 3 ]

e) Computing the derivative of $V$ along $f(x)$ yields:

$$\dot{V}(x) = \frac{\partial V}{\partial x} f(x) = -x_2^2(x_1^2 + x_2^2) \leq 0.$$

Notice that $\dot{V}$ is negative semidefinite. [ 2 ]

Answers

f) Clearly $\mathrm{Ker}[\dot{V}] = \{x : x_2 = 0\}$. To apply Lasalle's stability criterion we need to show that any invariant set inside $\mathrm{Ker}[\dot{V}]$ is in fact the origin. To this end taking a derivative of $x_2$ yields:

$$\{x : x_2 = 0 \text{ and } -x_2(x_1^2 + x_2^2) - x_1^5 = 0\} = \{0\}.$$

Hence, the Lasalle's criterion can be applied and the origin is globally asymptotically stable. [ 3 ]

g) The equilibria for $u \neq 0$ fulfill the equations:

$$x_2 = 0 \qquad -x_2(x_1^2 + x_2^2) - x_1^5 + u = 0$$

This yields a unique equilibrium: $x_e(u) = [\sqrt[5]{u}, 0]'$ for each constant value of $u$. It turns out that stability of this equilibrium can be studied by shifting coordinates to $\tilde{x} = x - x_e(u)$. The Lyapunov function needs to be modified accordingly. In particular, $\tilde{V}(x) = V(x) - x_1 u$ can be used as a Lyapunov function. [ 3 ]

Answers

3. Exercise

a) We consider the following scalar systems:

$$\Sigma_1: \qquad \dot{x}_1 = \text{atan}(-2x_1 + d_1),$$

$$\Sigma_2: \qquad \dot{x}_2 = (-x_2 + d_2)^3,$$

along with the interconnections $d_1 = x_2$ and $d_2 = x_1$ which define a feedback loop. [ 4 ]

b) We study ISS of $\Sigma_1$ by means of the Lyapunov function $V_1(x_1) = x_1^2/2$. This is positive definite and radially unbounded with class $\mathscr{K}_\infty$ lower and upper bounds $\alpha_1(|x_1|) = \alpha_2(|x_1|) = |x_1|^2/2$. Taking derivatives along solutions of $\Sigma_1$ yields:

$$\dot{V}_1 = x_1 \text{atan}(-2x_1 + d_1).$$

Consider first the case $x_1 \geq 0$. Choosing $|x_1| \geq |d_1|/(2-\varepsilon)$, for any $\varepsilon > 0$ yields:

$$|x_1| \geq |d_1|/(2-\varepsilon) \Rightarrow \dot{V}_1 = x_1 \text{atan}(-2x_1 + d_1) \leq x_1 \text{atan}(-2x_1 + |d_1|) \leq x_1 \text{atan}(-\varepsilon x_1).$$

A similar inequality holds when $x_1 \leq 0$. Notice that $x_1 \text{atan}(-\varepsilon x_1)$ is negative definite. Hence, choosing $\chi_1(r) := r/(2-\varepsilon)$ we see that $V_1$ is an ISS-Lyapunov function of type 2 as it fulfills the following implication:

$$|x_1| \geq \chi_1(|d_1|) \Rightarrow \dot{V}_1 \leq -\rho_1(|x_1|)$$

for some positive definite function $\rho_1$. Hence, $\Sigma_1$ is ISS with gain:

$$\gamma_1(r) = \alpha_1^{-1} \circ \alpha_2 \circ \chi_1(r) = \chi_1(r) = r/(2-\varepsilon).$$

[ 4 ]

c) We study ISS of $\Sigma_2$ by means of the Lyapunov function $V_2(x_2) = x_2^2/2$. This is positive definite and radially unbounded with class $\mathscr{K}_\infty$ the same lower and upper bounds $\alpha_1(|x_2|) = \alpha_2(|x_2|) = |x_2|^2/2$. Taking derivatives along solutions of $\Sigma_2$ yields:

$$\dot{V}_2 = x_2(-x_2 + d_2)^3.$$

Consider first the case $x_2 \geq 0$. Choosing $|x_2| \geq |d_2|/(1-\varepsilon)$, for any $\varepsilon > 0$ yields:

$$|x_2| \geq |d_2|/(1-\varepsilon) \Rightarrow \dot{V}_2 = x_2(-x_2 + d_2)^3 \leq x_2(-x_2 + |d_2|)^3 \leq x_2(-\varepsilon x_2)^3.$$

A similar inequality holds when $x_2 \leq 0$. Notice that $x_2(-\varepsilon x_2)^3$ is negative definite. Hence, choosing $\chi_2(r) := r/(1-\varepsilon)$ we see that $V_2$ is an ISS-Lyapunov function of type 2 as it fulfills the following implication:

$$|x_2| \geq \chi_2(|d_2|) \Rightarrow \dot{V}_2 \leq -\rho_2(|x_2|)$$

for some positive definite function $\rho_2$. Hence, $\Sigma_2$ is ISS with gain:

$$\gamma_2(r) = \alpha_1^{-1} \circ \alpha_2 \circ \chi_2(r) = \chi_2(r) = r/(1-\varepsilon).$$

[ 4 ]

Answers

d) We may conclude GAS at the origin of the overall system by virtue of the ISS Small Gain Theorem, in particular noticing that:

$$\gamma_1(\gamma_2(r)) = \frac{1}{(1-\varepsilon)(2-\varepsilon)} r < r$$

for all $\varepsilon$ sufficiently small and $r > 0$. This shows that the small-gain condition is satisfied.

[ 4 ]

e) Changing $+x_1$ with $-x_1$ in the equation for $\dot{x}_2$ will not affect the result. Indeed, the $\Sigma_2$ equation can be modified as:

$$\Sigma_2 : \qquad \dot{x}_2 = (-x_2 - d_2)^3,$$

On the other hand, the subsequent inequalities are still valid since:

$$|x_2| \geq |d_2|/(1-\varepsilon) \Rightarrow \dot{V}_2 = x_2(-x_2 - d_2)^3 \leq x_2(-x_2 + |d_2|)^3 \leq x_2(-\varepsilon x_2)^3.$$

[ 4 ]

Answers

4. Exercise

a) Computing derivatives of the output $y$ along solutions of the system yields:

$$\dot{y} = \dot{x}_1 - \dot{x}_2 = 2\sin(x_1 - x_2) + \sin(x_3 - x_1)$$

$$\begin{aligned}\ddot{y} &= 2\cos(x_1 - x_2)[\dot{x}_1 - \dot{x}_2] + \cos(x_3 - x_1)[\dot{x}_3 - \dot{x}_1] \\ &= 2\cos(x_1 - x_2)[2\sin(x_1 - x_2) + \sin(x_3 - x_1)] \\ &\quad + \cos(x_3 - x_1)[2\sin(x_1 - x_3) - \sin(x_2 - x_1) - x_3 - u]\end{aligned}$$

Notice that the coefficient of $u$ in $\ddot{y}$ equals $\cos(x_3 - x_1)$ which is different from 0 in a neighborhood of the origin. Hence, the relative degree is equal to 2. This is only locally defined as $\cos(x_3 - x_1) = 0$ whenever $x_3 - x_1 = \pi/2 + k\pi$. [ 3 ]

b) The input-output linearizing feedback is achieved by letting:

$$\begin{aligned}v &= 2\cos(x_1 - x_2)[2\sin(x_1 - x_2) + \sin(x_3 - x_1)] \\ &\quad + \cos(x_3 - x_1)[2\sin(x_1 - x_3) - \sin(x_2 - x_1) - x_3 - u].\end{aligned}$$

This gives the expression:

$$\begin{aligned}u = \frac{1}{\cos(x_3 - x_1)}\Big(&2\cos(x_1 - x_2)[2\sin(x_1 - x_2) + \sin(x_3 - x_1)] \\ &+ \cos(x_3 - x_1)[2\sin(x_1 - x_3) - \sin(x_2 - x_1) - x_3] - v\Big).\end{aligned}$$

[ 3 ]

c) We let $z_1 = y$, $z_2 = \dot{y}$ and $\xi = x_3 + x_1 - x_2$. This yields the following system of equations:

$$\begin{aligned}\dot{z}_1 &= z_2 \\ \dot{z}_2 &= v \\ \dot{\xi} &= \dot{x}_3 + \dot{x}_1 - \dot{x}_2 = 2\sin(x_1 - x_2) - x_3 = 2\sin(y) - \xi + y.\end{aligned}$$

Notice that the equation for $\dot{\xi}$ only involves $y$ and $\xi$. Therefore the $\xi$ dynamics are the so called internal dynamics. [ 6 ].

d) The internal dynamics are Input-to-State Stable. This can be seen by choosing the candidate Lyapunov function $V(\xi) = \xi^2$. Taking derivatives yields:

$$\dot{V} = 2\xi(-\xi + 2\sin(y) + y) \leq -2\xi^2 + 6|\xi||y| \leq -|\xi|^2 + 9|y|^2,$$

showing that $V$ is an ISS Lyapunov function of type 1 for the internal dynamics. [ 4 ]

e) We may locally asymptotically stabilize the system at the origin simply taking the feedback $v = -y - \dot{y}$. The stability is only local, since the original feedback applied to linearize the system was only defined locally in a neighborhood of 0. Moreover, the ISS of the internal dynamics guarantees that stabilizing the input-output equation yields an internally stable system (due to results on local asymptotic stability of cascaded stable systems). [ 2 ]

f) Replacing $-x_3$ with $+x_3$ in the last equation yields the following internal dynamics:

$$\dot{\xi} = 2\sin(x_1 - x_2) + x_3 = 2\sin(y) + \xi - y.$$

Notice that even for $y$ identically equal to 0 this is not locally asymptotically stable. Hence, using a stabilizing feedback for the $y$ dynamics does not result in a locally asymptotically stable closed-loop system at the origin. [ 2 ]

Answers